Synthetic Homotopy Theory

Exercises and Solutions for the

Midlands Graduate School (MGS)
13–17 April 2026, Nottingham, UK

by
Axel Ljungström and Aref Mohammadzadeh
School of Computer Science
University of Nottingham

Heavily based on notes from MGS 2024 by Ulrik Buchholtz and Mark Williams

NB Subject to change: Refresh if needed

Contents

1 Truncated and connected types **1**
1.1 Exercises 1

2 A menagerie of higher inductive types **4**
2.1 Exercises 4

3 The Hopf fibration and the LES **7**
3.1 Exercises 7

4 Blakers–Massey theorem and cohomology **9**
4.1 Exercises 9

Chapter 1

Truncated and connected types

1.1 Exercises

1. Show that for $n \geq m$ and any type A we have

$$\|\|A\|_n\|_m \simeq \|A\|_m.$$

2. Show that a type X is n-connected if and only if, for every n-type Y, the diagonal map

$$\delta : Y \to (X \to Y), \quad y \mapsto_\delta (x \mapsto y),$$

is an equivalence.

3. Show that the map $|_|_n : X \to \|X\|_n$ is n-connected.

4. Show that a type X is $(n+1)$-type if and only if the diagonal map of X,

$$\delta_X : X \to X \times X, \quad x \mapsto (x, x),$$

is n-truncated.

5. Consider a map $f : X \to Y$. We define its **n-image** to be the type $\mathrm{im}_n(f) := (y : Y) \times \|\mathrm{fib}_f(y)\|_n$.

   a) Show that we can factorise f through the n-image, that is, construct a commuting diagram:

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "p"'] & & Y \\ & \mathrm{im}_n(f) \arrow[ur, "i"'] & \end{tikzcd}$$

   b) Show that p is n-connected and i is n-truncated.

c) Show that if Z is any type, $g : X \to Z$ is n-connected, and $h : Z \to Y$ is n-truncated with $f = h \circ g$, then there is a unique equivalence $e : \mathrm{im}_n(f) \to Z$ with $e \circ p = g$ and $h \circ e = i$:

$$\begin{tikzcd}
 & Z \arrow[dr, "h"] \arrow[dd, dotted, "e"] & \\
X \arrow[ur, "g"] \arrow[rr, "f"] \arrow[dr, "p"'] & & Y \\
 & \mathrm{im}_n(f) \arrow[ur, "i"'] &
\end{tikzcd}$$

(*Hint*) It's easier if you assume g and h are projections of dependent sums.

6. (Eckmann–Hilton). All paths will take place in a fixed type X.

a) Given paths $p_0, p_1 : x = y$ and $q_0, q_1 : y = z$, and higher paths $r : p_0 = p_1$ and $s : q_0 = q_1$, define their **horizontal multiplication** $s * r : q_0 \cdot p_0 = q_1 \cdot p_1$ by path induction on r and p_0.

$$\begin{tikzcd}
x \arrow[r, bend left, "p_0"] \arrow[r, bend right, "p_1"'] \arrow[r, phantom, "\Downarrow r"] & y \arrow[r, bend left, "q_0"] \arrow[r, bend right, "q_1"'] \arrow[r, phantom, "\Downarrow s"] & z
\end{tikzcd}$$

b) Given paths fitting into the following diagram

$$\begin{tikzcd}
x \arrow[r, bend left, "p_0"] \arrow[r, bend right, "p_1"'] \arrow[r, phantom, "r_0\Downarrow"] \arrow[r, phantom, "r_1\Downarrow"] & y \arrow[r, bend left, "q_0"] \arrow[r, bend right, "q_1"'] \arrow[r, phantom, "\Downarrow s_0"] \arrow[r, phantom, "\Downarrow s_1"] & z
\end{tikzcd}$$

show we have the **interchange law**:

$$(s_1 * r_1) \cdot (s_0 * r_0) = (s_1 \cdot s_0) * (r_1 \cdot r_0)$$

c) Deduce that for $n \geq 2$, the path composition in $\Omega^n(X, x)$ is commutative, and thus that $\pi_n(X, x)$ is abelian.

7. (The Fundamental Theorem of Identity Types)

Let $R : X \to X \to \mathcal{U}$ be a reflexive binary relation. The following are equivalent

a) The obvious map[1] $x =_A y \to R(x, y)$ is an equivalence.

[1] i.e. the map defined by path induction which sends refl_x to the term of type $R(x, x)$ given to us by reflexivity of R

b) For any $x : X$, the total space $(y : X) \times (R(x, y))$ is contractible.

8. Recall that $\mathcal{U}^{\leq n} := (X : \mathcal{U}) \times \text{is-}n\text{-type}\ (X)$ is an $(n+1)$-type. This allows us to define $R : \|X\|_{n+1} \times \|X\|_{n+1} \to \mathcal{U}^{\leq n}$ by the truncation elimination:

$$R(|x|_{n+1}, |y|_{n+1}) := \|x = y\|_n$$

Apply the fundamental theorem of identity types with the above family to deduce that $(|x|_{n+1} =_{\|X\|_{n+1}} |y|_{n+1}) \simeq \|x = y\|_n$

9. Show that when X is connected, that $\pi_n(X, x_0) \cong \pi_n(X, x_1)$ are merely isomorphic for any $x_0, x_1 : X$, that is, construct an element of the type

$$(x_0, x_1 : X) \to \|\pi_n(X, x_0) \simeq \pi_n(X, x_1)\|.$$

10. We define the observational equality type family of the natural numbers $E : \mathbb{N} \to \mathbb{N} \to \mathcal{U}$ by

$$\begin{aligned} E(0, 0) &= 1 \\ E(\text{succ}(n), 0) &= 0 \\ E(0, \text{succ}(n)) &= 0 \\ E(\text{succ}(n), \text{succ}(m)) &= E(n, m) \end{aligned}$$

a) Define a natural map $n = m \to E(n, m)$ for all $n, m : \mathbb{N}$.

b) Use the fundamental theorem of identity types to show this natural map is an equivalence.

11. Show that $\|X\|_0$ satisfies the universal property of the set quotient of X modulo the equivalence relation $\sim$, where $x \sim x'$ if and only if $\|x =_X x'\|$.

Chapter 2

A menagerie of higher inductive types

2.1 Exercises

1. (Pushout-Pullback Duality) Consider a commutating square:

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "j"] \\ B \arrow[r, "i"'] & X \end{tikzcd}$$

   with $H : i \circ f \sim j \circ g$. Then the following are equivalent:

   (i) The square is a pushout square.

   (ii) the square

$$\begin{tikzcd} T^X \arrow[r, "-\circ j"] \arrow[d, "-\circ i"'] & T^C \arrow[d, "-\circ g"] \\ T^B \arrow[r, "-\circ f"'] & T^A \end{tikzcd}$$

   is a pullback square, for every type T.

2. Show that the two definitions of S^1 are equivalent, in two ways:

   (i) Show the HIT for S^1 is the same thing as $\Sigma 2$ by constructing an equivalence between them.

   (ii) By applying Exercise 2.1.

3. Show that evaluation at the loop gives pointed equivalence

$$(S^1 \to_\star X) \to_\star \Omega X$$

for any pointed type X.

Conclude, using the suspension–loop adjunction, that we have a natural equivalence

$$(S^n \to_\star X) \to_\star \Omega^n X$$

for all $n : \mathbb{N}$.

4. Show that a type X is a proposition iff $\mathrm{inl} : X \to X * X$ is an equivalence.

5. Show that $A \vee B \simeq A * B$ for propositions A, B, where $A \vee B := \|A + B\|$. (Beware the notational overload between the wedge sum and disjunction: It's not too bad in practice, because the only pointed proposition is the unit type!)

6. (i) Consider a span $A \xleftarrow{f} X \xrightarrow{g} B$. Show that the following square is pushout:

$$\begin{tikzcd} X + X \arrow[r, "{[\mathrm{id},\mathrm{id}]}"] \arrow[d, "f+g"'] \arrow[dr, phantom, "\ulcorner"] & X \arrow[d, "inr \circ g"] \\ A + B \arrow[r, "{[\mathrm{inl},\mathrm{inr}]}"'] & A \amalg_X B \end{tikzcd}$$

   (ii) Apply this to show that for every type X we get $\Sigma X \simeq 2 * X$.

7. (Smash Product - Pointed Map Space Adjunction)

   (i) Prove that for each pointed type B, the fucnction $(-) \wedge B$ is the left adjoint to the function $B \to_* (-)$,i.e. for every pointed types A and C, there is a pointed equivalence:

$$A \to_* (B \to_* C) \simeq_* (A \wedge B) \to_* C$$

   (ii) Use this to prove that $S^1 \wedge X \simeq \Sigma X$ for each type X.

8. Show that the smash product is equivalently the pushout:

$$\begin{tikzcd} X + Y \arrow[r] \arrow[d] & X \times Y \arrow[d] \\ 1 + 1 \arrow[r] & X \wedge Y \end{tikzcd}$$

9. (i) For all pointed types X, Y, and A, prove there is a pointed equivalence:

$$X * Y \to_* A \simeq_* X \to (Y \to_* \Omega A)$$

   (ii) For pointed types X, Y prove that

$$X * Y \simeq_* \Sigma(X \wedge Y).$$

10. We can define the unit interval type $\mathbb{I}$ to be the pushout of the constant span $1 \leftarrow 1 \rightarrow 1$. Working in Martin-Löf type theory *without* univalence (and function extensionality) but with pushouts with strict definitional computation rules, prove that

    (i) $[((x, y) : A \times A) \times (x = y)] \simeq (\mathbb{I} \rightarrow A)$

    (ii) Use the above equivalence (or, if you prefer, a variation thereof) to derive function extensionality.

11. Show that the square

$$\begin{tikzcd} S^1 \arrow[r] \arrow[d] & 1 \arrow[d] \\ S^1 \times S^1 \arrow[r] & S^2 \vee S^1 \end{tikzcd}$$

is a pushout.

Chapter 3

The Hopf fibration and the long exact sequence

3.1 Exercises

1. Construct the multiplication on the circle $\mu : S^1 \to (S^1 \to S^1)$ by circle induction. Show that for each $z : S^1$ the map $\mu(z, -) : S^1 \to S^1$ is an equivalence.

2. Construct a retraction $\rho : \Omega S^2 \to S^1$ of $\eta : S^1 \to \Omega S^2$. (*Hint*) Use the Hopf fibration.

3. (i) Show that if the square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "j"] \\ B \arrow[r, "i"'] & X \end{tikzcd}$$

is a pullback, then so is

$$\begin{tikzcd} \Omega A \arrow[r, "\Omega g"] \arrow[d, "\Omega f"'] & \Omega C \arrow[d, "\Omega j"] \\ \Omega B \arrow[r, "\Omega i"'] & \Omega X \end{tikzcd}$$

(ii) For a map $A \xrightarrow{f} B$ show that $\Omega\,(\mathrm{fib}\ f) \simeq \mathrm{fib}\,(\Omega f)$

(iii) Show that if $A \xrightarrow{f} B \xrightarrow{g} X$ is a fiber sequence, then $\Omega A \xrightarrow{\Omega f} \Omega B \xrightarrow{\Omega g} \Omega X$ is a fiber sequence.

4. (Flattening Lemma) Given a span $A \xleftarrow{f} C \xrightarrow{g} B$ with pushout D and a type family $P : D \to \mathcal{U}$ defined by

$$\begin{aligned} P_{\text{inl}} &: A \to \mathcal{U} \\ P_{\text{inr}} &: B \to \mathcal{U} \\ P_{\text{glue}} &: (c : C) \to P_{\text{inl}}(f(c)) \simeq P_{\text{inr}}(g(c)), \end{aligned}$$

the total space of P is equivalent to the pushout of the span

$$\big((a : A) \times P_{\text{inl}}(a)\big) \longleftarrow \big((c : C) \times P_{\text{inl}}(f(c))\big) \longrightarrow \big((b : B) \times P_{\text{inr}}(b)\big),$$

where the left maps is $(c, u) \mapsto (f(c), u)$ and the right map is $(c, u) \mapsto (g(c), P_{\text{glue}}(c)(u))$.

5. (i) Find a pointed type X and a map $S^2 \xrightarrow{f} X$ such that the fiber of f is S^3.

   (ii) Is there any pointed map out of S^1 such that its fiber is S^2?

6. (Splitting Lemma) For a map $E \xrightarrow{r'} B$, Consider the induced map $\Omega E \xrightarrow{r := \Omega r'} \Omega B$. Prove that if r has a section, then:

$$\Omega E = \Omega B \times \text{fib}_r$$

Chapter 4

Blakers–Massey theorem and cohomology

4.1 Exercises

1. (Join Connectivity Lemma) Prove that if X is n-connected and Y is m-connected, then $X * Y$ is $(n+m+2)$-connected.

2. (Dual Blakers–Massey Theorem) In any pullback square of maps f and g,

$$\begin{tikzcd} X \times_Z Y \arrow[r, "\mathrm{snd}"] \arrow[d, "\mathrm{fst}"'] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z, \end{tikzcd}$$

   where f is n-connected and g is m-connected, if we form the pushout Q of the span $X \leftarrow X \times_Z Y \rightarrow Y$, then we get an induced map (the cogap) $c : Q \rightarrow Z$. Prove that this is $(n+m+2)$-connected. Hint: use the Join Connectivity Lemma by identifying the fibers of the cogap map with the joins of the fibers of legs in the cospan.

3. (Another proof for the Wedge Connectivity Lemma) Deduce the Wedge Connectivity Lemma from the dual Blakers–Massey, i.e., consider the pullback square, for pointed types X, Y,

$$\begin{tikzcd} 1 \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d] \\ X \arrow[r] & X \times Y. \end{tikzcd}$$

   Show that if X is n-connected and Y is m-connected, then the wedge inclusion map $X \vee Y \rightarrow X \times Y$ is $(n+m)$-connected.

4. Show that the counit $\varepsilon : \Sigma\Omega X \rightarrow X$ of the loop–suspension adjunction is $2n$-connected if X is n-connected.

5. For the following pushout square:

$$\begin{tikzcd} R \arrow[r, "g"] \arrow[d, "f"'] & B \arrow[d, "\mathrm{inr}"] \\ A \arrow[r, "\mathrm{inl}"'] \arrow[ur, phantom, "\ulcorner"] & Q \end{tikzcd}$$

   (i) Show that f is surjective if and only if inr is.

   (ii) Show that the pushout of an embedding is an embedding, i.e., if f is (-1)-truncated, then so is inr in the pushout square.(Hint: Use flattening lemma stated in exercise 3.4)

6. For all n and pointed types X, Y prove that:

$$(X \times Y \to_* K_n) \simeq (X \to_* K_n) \times (Y \to_* K_n) \times (X \wedge Y \to_* K_n)$$

   Hint : Use pushout-pullback duality (exercise 2.1) and splitting lemma (exercise 3.6).

7. Use the previous exercise to compute all cohomology groups of the torus $S^1 \times S^1$.

8. For a type X with contractbile suspension ($\Sigma X \simeq 1$) show that all reduced cohomology groups vanish.